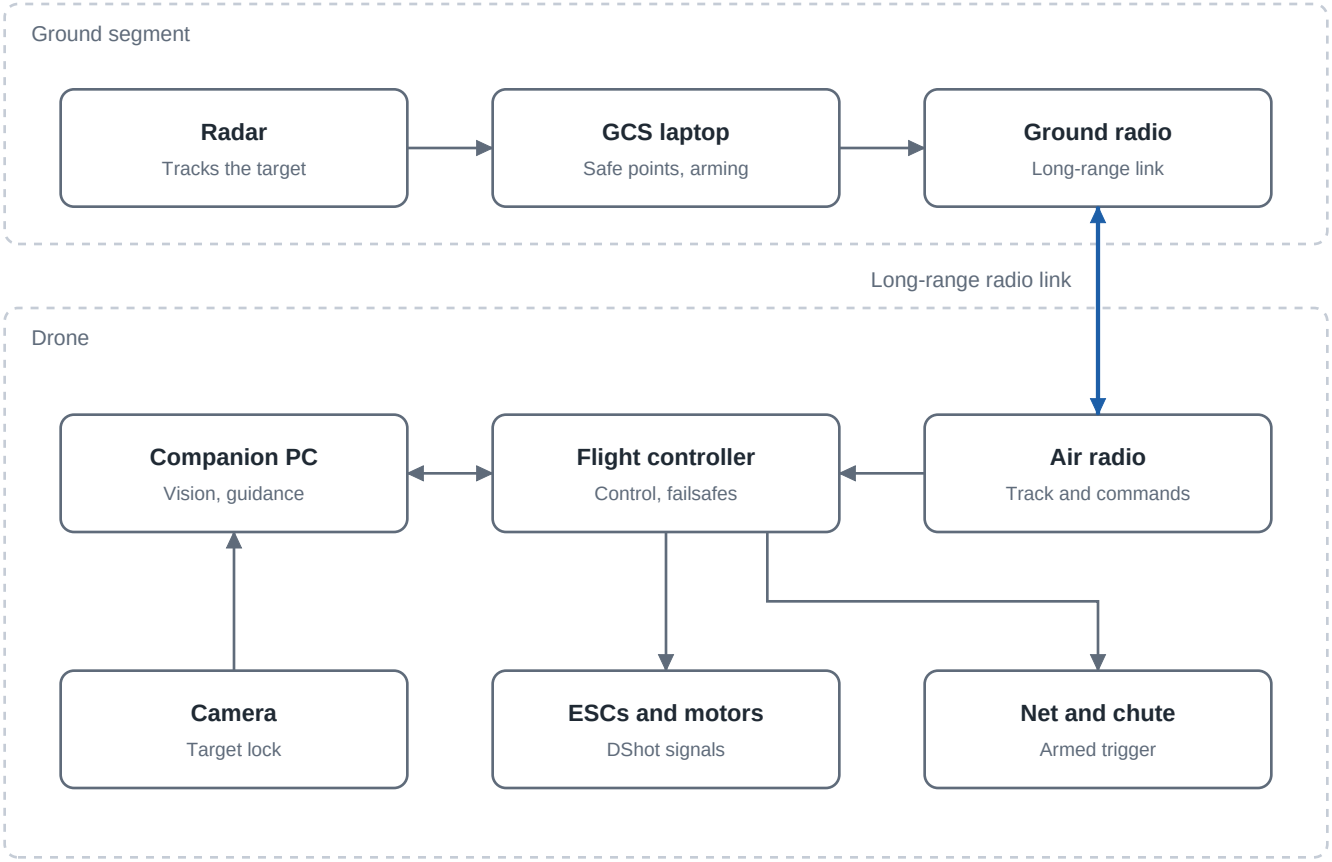
SKYFORCE Interceptor: Communications, Compute and Wiring Architecture

Team 08, Anti-Drone Interceptor, Senior Track. Prepared 10 October 2026.

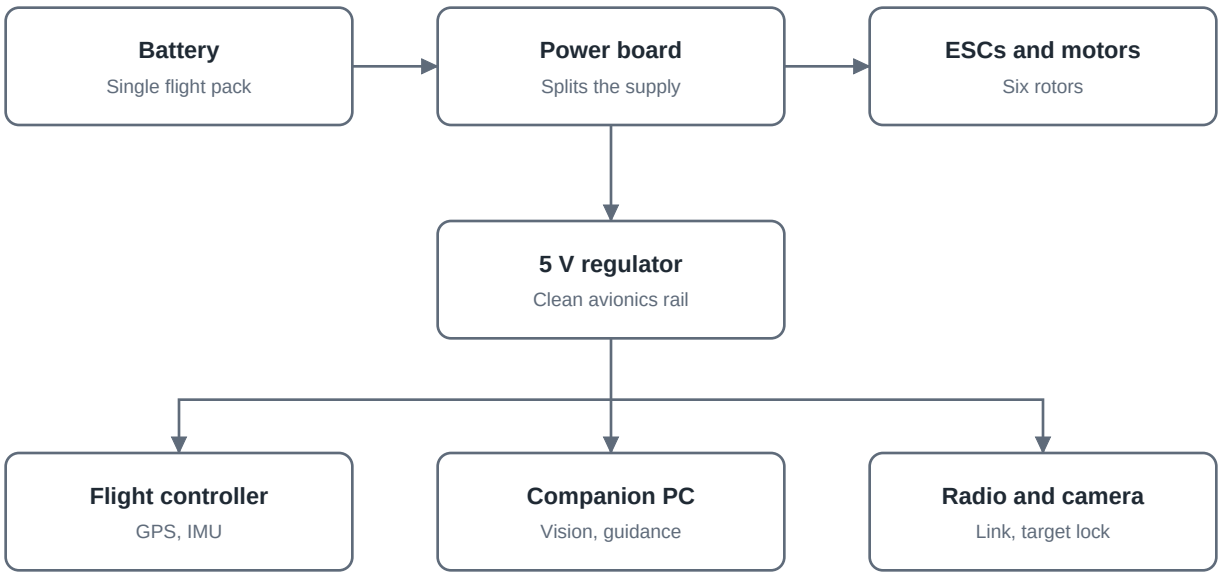
These are working design choices, not verified product specifications. No radio, radar, companion computer or camera model has been selected yet, and the link budget uses assumed values. The mass, power and dimension numbers are in the separate sizing document.

Block diagram 1: signal and control path



The radar feeds the ground control (GCS) laptop. The laptop sends tracks and commands over one long-range radio link to the drone, where the flight controller and companion computer act on them.

Block diagram 2: power distribution



One battery feeds a power board. The board supplies the ESCs directly and the avionics through a separate regulated 5 V rail, so motor noise cannot reset sensitive electronics.

1. How the radar reaches the drone

The radar never talks to the drone directly.

- The ground radar detects and tracks the target and sends tracks to the GCS laptop over a wired link (Ethernet or serial, depending on the radar).
- The GCS converts the track into a target position and velocity in the drone frame, using the radar surveyed GPS position and orientation, and plans the intercept.
- The GCS sends the target track and commands to the drone over the long-range radio link.
- The drone flies to the cued point and hands over to its camera for terminal guidance.

This matches the radar-cued approach used by the Fortem F700. The radar track update rate sets how often the cue refreshes (for the TrueView R30 it is configurable from 64 ms to 3 s).

2. Long-range link (1 km and beyond)

Standard MAVLink radios in the 433 or 900 MHz range, with a decent antenna and line of sight, normally reach well beyond 1 km, so this requirement does not need an exotic link. The baseline is a 1 W class long-range MAVLink radio on the drone and on the GCS. Check the legal band and power for India (the licence-free 865 to 867 MHz band, and 2.4 and 5.8 GHz) before choosing the model.

```
FSPL (dB) = 20 log10(d_km) + 20 log10(f_MHz) + 32.44          (free space, about 866 MHz)
At 1 km:  0     + 58.75 + 32.44 = 91 dB
At 4 km:  12.04 + 58.75 + 32.44 = 103 dB

Assumed: Tx 30 dBm (1 W), 2 dBi antenna each end, receiver sensitivity -110 dBm
Margin at 1 km = 30 + 2 + 2 - 91  + 110 = 53 dB
Margin at 4 km = 30 + 2 + 2 - 103 + 110 = 41 dB
```

The margin leaves room for fading, antenna orientation and obstacles. Replace the assumed values with the chosen radio datasheet and run a real range test.

Add an independent RC link (2.4 GHz class) to the flight controller so a safety pilot can take over manually. A video downlink is optional, because terminal guidance runs on board.

3. Companion computer and flight controller

- The companion computer connects to the flight controller over a serial (UART) port using MAVLink 2. With ROS2, MAVROS or MAVSDK provide the bridge.
- The companion sends velocity and yaw-rate setpoints in guided mode. The flight controller keeps the low-level control loops, failsafes and geofence.
- The camera connects directly to the companion computer (USB or MIPI CSI).

From	To	Interface
Radar	GCS laptop	Ethernet or serial (radar dependent)
GCS laptop	Ground radio	USB serial, MAVLink
Air radio	Flight controller	UART, MAVLink
Companion computer	Flight controller	UART, MAVLink 2
Camera	Companion computer	USB or MIPI CSI
GPS and IMU	Flight controller	UART and I2C
Flight controller	ESCs	DShot
Flight controller	Net and parachute trigger	PWM or relay output, through an arming interlock
RC receiver	Flight controller	SBUS or CRSF

4. Onboard software runs offline

The companion computer needs no internet. Everything runs on the drone from local code and locally stored model files:

- Camera frames come in.
- The detector or tracker locks on to the target, starting from the radar cue.
- The guidance law (for example proportional navigation or visual servoing) converts the target image position into velocity setpoints.
- The flight controller flies them.
- When the target is inside the net range window with a stable lock, the net trigger is requested.

A classical OpenCV tracker or a small detector is enough. Proposed firing rules (team to confirm):

- The operator must give an explicit arm command from the GCS.
- The flight controller state must be acceptable (geofence, battery, link).
- Range and lock confidence must be inside the design window (5 to 15 m, a design target).
- A hardware arming switch or interlock sits in the trigger circuit.
- Fire permission expires if the link heartbeat is lost.

5. Camera and landing decision

One forward gimbal camera is used for target lock. There is no second, downward camera: once the target is netted directly beneath the drone, a downward view would be blocked. Landing safety is handled with GPS safe points instead.

- Safe points are set as GPS coordinates before launch and loaded into the GCS and the flight controller.
- After capture, the drone compares its GPS position with the safety area. If it is inside, it releases the parachute there.

- If it is outside, it tows the netted target to the nearest safe point and releases the parachute there.
- The tow energy check in the sizing document assumes a 60 s tow, so safe points must be placed within that tow time of the engagement area.
- **Open item:** decide what the drone does if no safe point is reachable on the remaining battery.
- A parachute on the net or target was considered and not chosen: the target would drift with the wind and the landing point is harder to control.

What the camera sees (1920 pixels across a 90 degree field, 0.8 m target span, an assumed upper span for a 5 kg drone):

```
pixels per degree = 1920 / 90 = 21.3
At 100 m: 0.8 / 100 rad = 0.458 deg = about 10 pixels
At  30 m: 1.53 deg = about 33 pixels
At  15 m: 3.06 deg = about 65 pixels
```

The target is only about 10 pixels at 100 m, so the radar cue must bring the drone close before the camera can lock on, or a narrower lens should be used.

6. Power wiring

- The battery feeds a power board. The board feeds the six ESCs directly and a separate 5 V regulator for the flight controller, companion computer, radio and camera. Keep avionics on their own regulated rail.
- Peak pack current at sprint is about 81 A, or about 14 A per motor (see the sizing document).
- Wire sizing rule of thumb (verify against connector and wire datasheets): about 8 AWG and an XT90-class connector for the main bus, and 14 to 16 AWG for each ESC lead.
- The net and parachute trigger circuit sits behind its own arming switch so it cannot fire from a software fault alone.